

**MARDIE, Australia**WMO: **943060**Lat: **21.191S**Lon: **115.980E**Elev: **39**StdP: **14.68**Time Zone: **8.00 (E08)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 50.5       | 52.8 | 16.6                          | 12.7 | 80.1 | 21.3 | 16.0 | 78.9 | 24.7                  | 74.5 | 22.6 | 75.7 | 7.3                      | 120  | 0.465 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 17.4                   | 105.8           | 70.6 | 102.8 | 71.2 | 100.3 | 71.6 | 82.1                | 86.4 | 80.9 | 86.9 | 80.0 | 87.1 | 13.0                 | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.4                            | 163.4 | 83.2 | 79.8 | 154.8 | 83.0 | 78.3 | 147.2 | 83.3 | 46.1          | 86.2 | 44.8 | 86.5 | 43.7 | 87.2 | 86.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 23.6              | 21.2 | 19.4 | DB                         | 46.5 | 112.6              | 1.7 | 3.6                                                | 45.3 | 115.2      | 44.3 | 117.3      | 43.3 | 119.3      | 42.1 | 122.0 |  |
|                   |      |      | WB                         | 39.0 | 83.7               | 2.5 | 1.9                                                | 37.1 | 85.0       | 35.7 | 86.1       | 34.2 | 87.2       | 32.4 | 88.6  |  |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 79.2   | 87.4  | 87.6  | 87.6  | 83.5  | 75.5  | 69.7  | 68.4  | 69.7  | 73.5  | 79.9  | 82.3  | 85.9  |
|                                                                               | DBStd       | 8.32   | 4.51  | 4.02  | 3.84  | 3.88  | 4.20  | 4.16  | 3.67  | 3.16  | 3.63  | 4.96  | 4.62  | 4.73  |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | HDD65       | 23     | 0     | 0     | 0     | 0     | 0     | 8     | 12    | 3     | 0     | 0     | 0     | 0     |
|                                                                               | CDD50       | 10662  | 1159  | 1054  | 1165  | 1005  | 790   | 591   | 571   | 610   | 706   | 927   | 970   | 1114  |
|                                                                               | CDD65       | 5211   | 694   | 634   | 700   | 555   | 325   | 150   | 118   | 148   | 256   | 462   | 520   | 649   |
|                                                                               | CDH74       | 63912  | 9375  | 8680  | 9570  | 6661  | 3066  | 1220  | 1103  | 1613  | 2711  | 5357  | 6093  | 8463  |
| CDH80                                                                         | 33291       | 5195   | 4863  | 5430  | 3563  | 1296  | 286   | 241   | 460   | 1162  | 2887  | 3227  | 4681  |       |
| Wind                                                                          | WSAvg       | 9.4    | 10.5  | 9.8   | 8.9   | 7.7   | 7.9   | 8.7   | 8.4   | 8.9   | 9.4   | 10.1  | 11.1  | 10.7  |
| Precipitation                                                                 | PrecAvg     | 12.7   | 1.6   | 3.2   | 2.0   | 0.7   | 1.7   | 1.9   | 0.5   | 0.3   | 0.0   | 0.0   | 0.0   | 0.5   |
|                                                                               | PrecMax     | 33.7   | 9.5   | 26.6  | 11.8  | 5.7   | 8.3   | 9.3   | 3.6   | 2.2   | 0.4   | 0.5   | 0.4   | 6.7   |
|                                                                               | PrecMin     | 3.3    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                               | PrecStd     | 6.9    | 2.4   | 4.4   | 2.9   | 1.3   | 2.1   | 2.3   | 0.7   | 0.5   | 0.1   | 0.1   | 0.1   | 1.2   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 110.6 | 108.2 | 107.1 | 103.1 | 96.2  | 88.1  | 87.9  | 90.1  | 97.4  | 103.9 | 106.6 | 108.5 |
|                                                                               |             | MCWB   | 69.6  | 73.2  | 71.0  | 68.8  | 64.2  | 62.8  | 60.4  | 58.7  | 63.1  | 65.1  | 66.6  | 69.4  |
|                                                                               | 2%          | DB     | 104.0 | 103.4 | 103.0 | 99.9  | 93.0  | 85.7  | 85.2  | 87.2  | 93.1  | 100.1 | 101.7 | 103.7 |
|                                                                               |             | MCWB   | 73.6  | 73.7  | 72.5  | 68.0  | 63.8  | 61.2  | 59.9  | 59.3  | 62.4  | 65.4  | 67.9  | 70.3  |
|                                                                               | 5%          | DB     | 99.8  | 99.7  | 99.9  | 97.4  | 90.1  | 83.4  | 82.7  | 84.7  | 89.9  | 96.9  | 97.7  | 99.8  |
|                                                                               |             | MCWB   | 75.1  | 75.2  | 73.5  | 68.1  | 63.6  | 60.7  | 59.1  | 59.0  | 62.0  | 66.3  | 68.4  | 71.4  |
|                                                                               | 10%         | DB     | 96.4  | 96.6  | 97.1  | 94.7  | 87.3  | 81.1  | 80.2  | 82.5  | 86.8  | 93.6  | 94.1  | 96.5  |
|                                                                               |             | MCWB   | 75.8  | 76.3  | 73.7  | 68.4  | 63.5  | 60.3  | 59.0  | 58.9  | 61.9  | 65.7  | 68.7  | 72.6  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 83.5  | 83.8  | 82.7  | 81.9  | 77.8  | 75.5  | 70.8  | 70.4  | 71.5  | 74.9  | 77.5  | 81.8  |
|                                                                               |             | MCDB   | 87.0  | 86.7  | 85.1  | 83.8  | 79.3  | 76.8  | 76.0  | 75.9  | 78.4  | 82.9  | 83.6  | 85.4  |
|                                                                               | 2%          | WB     | 81.7  | 82.0  | 81.2  | 78.6  | 74.3  | 72.0  | 67.5  | 67.4  | 68.9  | 72.7  | 76.0  | 79.9  |
|                                                                               |             | MCDB   | 86.5  | 88.0  | 85.9  | 84.4  | 79.2  | 75.7  | 73.4  | 75.4  | 78.7  | 83.6  | 84.5  | 88.0  |
|                                                                               | 5%          | WB     | 80.5  | 80.8  | 80.0  | 76.9  | 72.0  | 69.1  | 65.3  | 65.4  | 67.4  | 71.3  | 74.6  | 78.3  |
|                                                                               |             | MCDB   | 86.9  | 87.9  | 86.5  | 84.2  | 78.5  | 74.6  | 73.9  | 75.1  | 78.5  | 83.7  | 84.5  | 87.2  |
|                                                                               | 10%         | WB     | 79.6  | 79.7  | 79.0  | 75.3  | 70.2  | 66.7  | 63.4  | 63.4  | 66.1  | 70.1  | 73.2  | 77.0  |
|                                                                               |             | MCDB   | 87.3  | 87.8  | 86.9  | 83.7  | 77.9  | 73.7  | 73.8  | 74.7  | 78.0  | 83.8  | 84.8  | 87.0  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 17.1  | 17.4  | 18.9  | 22.6  | 23.4  | 22.9  | 25.0  | 27.1  | 27.4  | 26.5  | 22.8  | 19.8  |
|                                                                               |             | MCDBR  | 23.4  | 22.9  | 23.8  | 26.2  | 26.9  | 25.3  | 28.1  | 29.9  | 31.9  | 30.7  | 28.5  | 26.2  |
|                                                                               |             | MCWBR  | 9.2   | 8.4   | 9.4   | 10.6  | 9.9   | 10.0  | 11.2  | 11.7  | 12.5  | 12.3  | 11.5  | 10.4  |
|                                                                               | 5% WB       | MCDBR  | 16.3  | 17.8  | 18.6  | 19.8  | 19.0  | 15.8  | 21.1  | 23.8  | 26.5  | 25.7  | 22.6  | 18.8  |
|                                                                               |             | MCWBR  | 7.0   | 7.1   | 8.0   | 9.4   | 8.9   | 8.7   | 10.4  | 11.8  | 11.9  | 11.5  | 9.5   | 7.9   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.400 | 0.396 | 0.378 | 0.354 | 0.333 | 0.322 | 0.314 | 0.317 | 0.333 | 0.352 | 0.357 | 0.374 |
|                                                                               | taud        |        | 2.497 | 2.525 | 2.562 | 2.588 | 2.582 | 2.594 | 2.595 | 2.544 | 2.499 | 2.471 | 2.494 | 2.520 |
|                                                                               | Ebn at Noon |        | 300   | 297   | 295   | 289   | 284   | 281   | 287   | 297   | 304   | 307   | 311   | 308   |
|                                                                               | Edh at Noon |        | 37    | 35    | 33    | 30    | 27    | 26    | 27    | 30    | 34    | 37    | 37    | 36    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 2400  | 2284  | 2031  | 1746  | 1466  | 1323  | 1433  | 1780  | 2118  | 2395  | 2567  | 2575  |
|                                                                               | RadStd      |        | 133   | 148   | 137   | 99    | 114   | 96    | 99    | 55    | 76    | 65    | 70    | 108   |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page